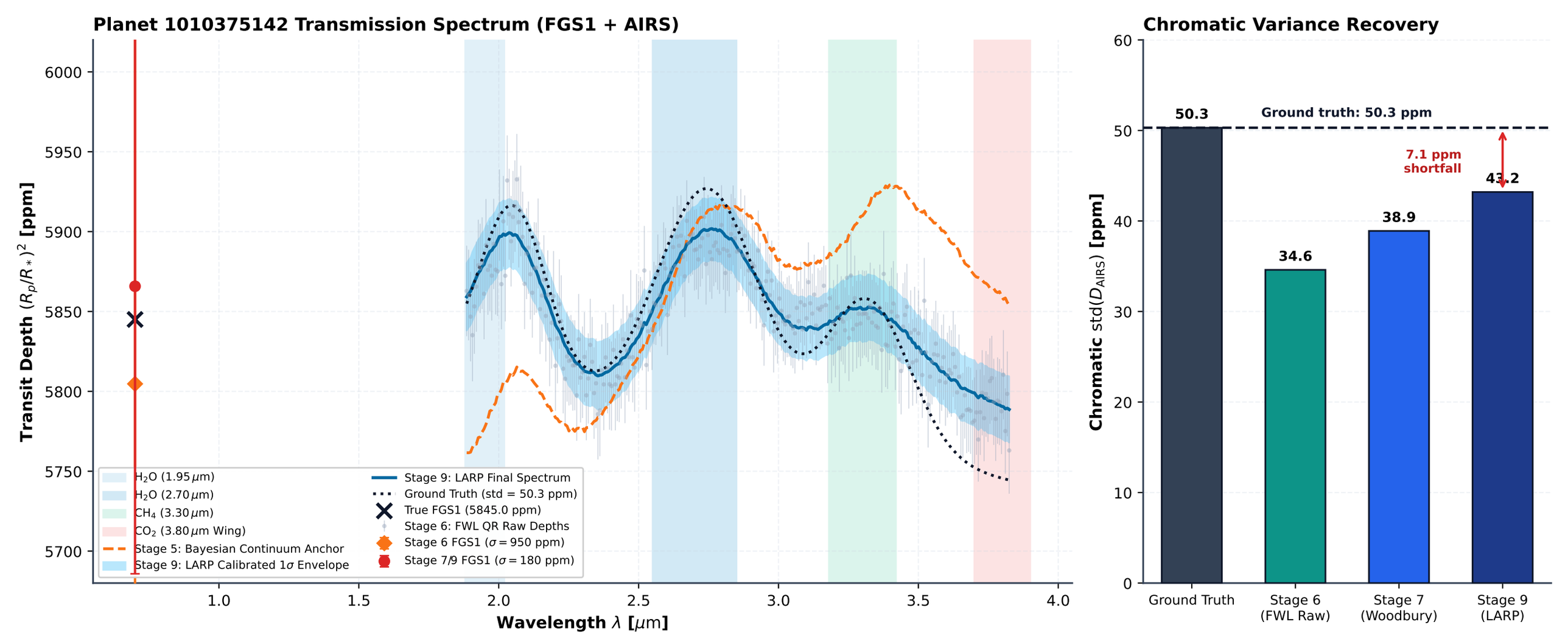

Planet 1010375142 Transmission Spectrum (FGS1 + AIRS)
Transit Depth $(R_p/R_*)^2$ [ppm]
Wavelength λ [μm]
6000
5950
5900
5850
5800
5750
5700
1.0
1.5
2.0
2.5
3.0
3.5
4.0
H_2O (1.95 μm)
H_2O (2.70 μm)
CH_4 (3.30 μm)
CO_2 (3.80 μm Wing)
Stage 5: Bayesian Continuum Anchor
Stage 9: LARP Calibrated 1σ Envelope
Stage 9: LARP Final Spectrum
Ground Truth (std = 50.3 ppm)
True FGS1 (5845.0 ppm)
Stage 6: FWL QR Raw Depths
Stage 6 FGS1 (σ = 950 ppm)
Stage 7/9 FGS1 (σ = 180 ppm)
Chromatic Variance Recovery
Chromatic std(D_{AIRS}) [ppm]
60
50
40
30
20
10
0
50.3
Ground truth: 50.3 ppm
7.1 ppm
shortfall
34.6
38.9
43.2
Ground Truth
Stage 6
(FWL Raw)
Stage 7
(Woodbury)
Stage 9
(LARP)